

Question

The chemical formula of salt **A** can be represented as $M(XY)_2$, where the anion is highly toxic. The mass fraction of **M** in **A** is 53.10%, and both **X** and **Y** are second-period elements.

Dissolving **A** in a potassium salt solution with the same anion forms a red solution. Boiling it in air causes an oxidation reaction to produce a solution of yellow salt **B**. Reacting **B** with hydrochloric acid and diethyl ether yields colorless crystals **C**. **C** reacts with nitromethane in methanol to obtain **D**. Neglecting water of crystallization, the molar mass of **D** is 42.08 g/mol greater than that of **C**.

Determine the correctness of the conclusions regarding the above content and select the option corresponding to all the correct conclusions.

Conclusion 1: The element **M** can react directly with carbon monoxide to form a mononuclear carbonyl complex.

Conclusion 2: Theoretical prediction is that **B** should be diamagnetic.

Conclusion 3: Compound **D** contains two ligands in a 1:1 ratio.

Conclusion 4: There is only one type of atom coordinating to **M** in compound **D**.

Conclusion 5: **C** exhibits a weak acidity comparable to that of hydrocyanic acid in aqueous solution.

A. All other options are incorrect

B. Conclusion 1, Conclusion 2, Conclusion 3

C. Conclusion 1, Conclusion 2, Conclusion 4

D. Conclusion 1, Conclusion 2, Conclusion 5

E. Conclusion 1, Conclusion 3, Conclusion 4

F. Conclusion 1, Conclusion 3, Conclusion 5

G. Conclusion 1, Conclusion 4, Conclusion 5

H. Conclusion 2, Conclusion 3, Conclusion 4

I. Conclusion 2, Conclusion 3, Conclusion 5

J. Conclusion 2, Conclusion 4, Conclusion 5

K. Conclusion 3, Conclusion 4, Conclusion 5

Answer

Correct Answer: **H**

Detailed Explanation

The highly toxic ion composed of only second-period elements in a 1:1 ratio is CN^- , so we can write the equation:

$$\frac{M(\mathbf{M})}{M(\mathbf{M}) + 2 \times (12.01 + 14.01)} = 53.10\%$$

Solving for $M(\mathbf{M}) = 58.9(\text{g/mol})$, corresponding to cobalt or nickel. However, nickel cyanide complexes cannot be oxidized by air, which does not meet the requirements of the question. Therefore, **M** is cobalt, and **A** is $Co(CN)_2$.

CHECKPOINT

2 PTS

M is cobalt, and **A** is $Co(CN)_2$

Cobalt reacts directly with carbon monoxide to only produce $Co_2(CO)_8$, which is not a mononuclear carbonyl complex. Conclusion 1 is incorrect.

CHECKPOINT

1 PTS

Cobalt reacts directly with carbon monoxide to only produce $Co_2(CO)_8$, which is not a mononuclear carbonyl complex.

When $Co(CN)_2$ is dissolved in potassium cyanide solution, Co is coordinated to form the deep red $[Co(CN)_6]^{4-}$. Boiling the solution in air, $[Co(CN)_6]^{4-}$ is oxidized to the yellow $[Co(CN)_6]^{3-}$. Therefore, **B** is $[Co(CN)_6]^{3-}$.

CHECKPOINT

1 PTS

B is $[Co(CN)_6]^{3-}$

CN^- is a strong field ligand, and Co^{3+} has a high valence, resulting in a large splitting energy. Therefore, $[Co(CN)_6]^{3-}$ is a low-spin complex, where Co has a d^6 electron configuration, which in an octahedral field is a $(t_{2g})^6(e_g)^0$ electron configuration. All electrons are paired, so the complex is predicted to be diamagnetic. Conclusion 2 is correct.

CHECKPOINT

1 PTS

$[Co(CN)_6]^{3-}$ has a $(t_{2g})^6(e_g)^0$ electron configuration in an octahedral field, all electrons are paired, and it is diamagnetic

B is treated with hydrochloric acid and diethyl ether to obtain **C**. Since complexes formed by +3 valent cobalt have kinetic stability, it can be inferred that **C** is likely the acid or acid hydrate corresponding to the anion, where the complex anion remains unchanged, still being $[Co(CN)_6]^{3-}$. Therefore, it is speculated that **C** is

$H_3[Co(CN)_6] \cdot nH_2O$, where n represents the number of crystal water molecules, which cannot be deduced from the existing conditions.

CHECKPOINT

1 PTS

C is $H_3[Co(CN)_6] \cdot nH_2O$

Ignoring crystal water, the molar mass of compound **D** is 42.08 g/mol more than that of **C**, which corresponds precisely to 3 methylene groups. The possible reactions of nitromethane include oxidation and substitution reactions. Here, *Co* has already been oxidized to the stable +3 valence, so a substitution reaction should have occurred. Note that the *N* atom in $[Co(CN)_6]^{3-}$ has a lone pair of electrons and is nucleophilic. Therefore, it is speculated that the *N* atom undergoes an S_N2 reaction by nucleophilic attack on nitromethane. Based on the molar mass, it can be deduced that **D** is $[Co(CN)_3(CNCH_3)_3]$.

CHECKPOINT

1 PTS

D is $[Co(CN)_3(CNCH_3)_3]$

D contains cyanide and methyl isocyanide ligands in a 1:1 ratio. Conclusion 3 is correct.

Since complexes formed by +3 valent cobalt have kinetic stability, it can be inferred that the coordination of the *C* atom to *Co* will not change in the above nucleophilic substitution reaction. In **D**, only the *C* atom is still coordinated to cobalt. Conclusion 4 is correct.

CHECKPOINT

1 PTS

Only the *C* atom is still coordinated to cobalt in **D**.

Since the basicity of the cyanide coordinated to cobalt is greatly reduced, its ability to bind to protons is reduced. Therefore, the acidity of **C** is much stronger than that of hydrocyanic acid, and it is a strong acid in aqueous solution. Conclusion 5 is incorrect.

CHECKPOINT

1 PTS

The acidity of **C** is much stronger than that of hydrocyanic acid

In summary, choose option H.